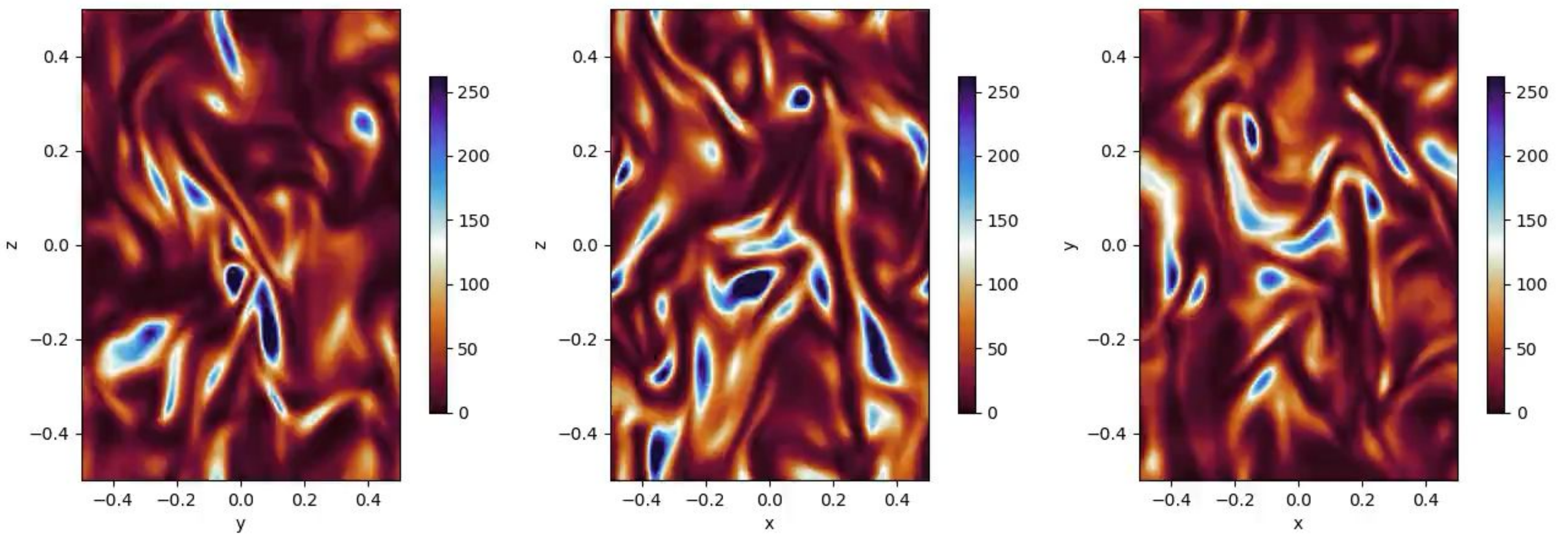
SOLVE THE FOLLOWING PROBLEM

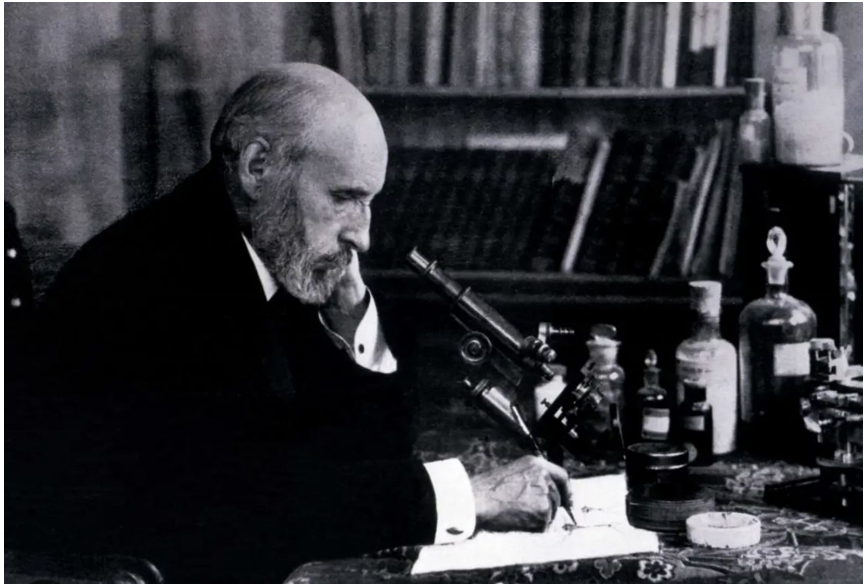
You are given multiple 2D time-dependent simulation datasets stored as (time, x, y) arrays. Analyze the evolution of each dataset, identify patterns or structures in space and time, compare across different initial conditions (which are 4 corresponding to each file), and summarize your findings about the underlying dynamics.

Propose a Partial Differential Equation that describes the dynamics of the physical system.



The **Founding Fathers** of (understanding) Human and (inventing) Artificial Intelligence were Spanish

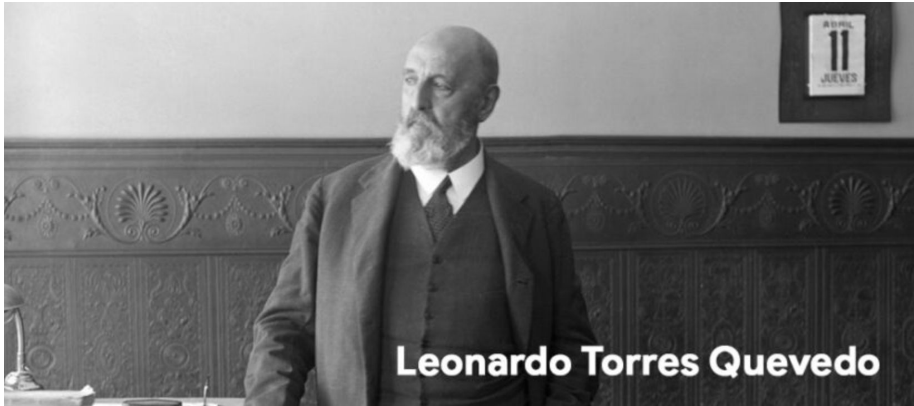
Santiago Ramón y Cajal



One of **Torres-Quevedo's** most surprising contributions was in the field of **artificial intelligence**. His most notable invention in this area was the "**Ajedrecista**," considered one of the first game-playing automata in history.

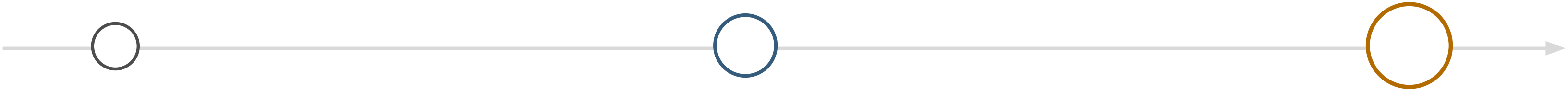
The Ajedrecista, first presented in 1912, was capable of playing a chess endgame (king and rook versus king) **completely automatically.**

Although the rules were limited, the device could "think" several moves ahead and execute strategies to win the game.



AI in science — and why cosmology is a stress test

Panel framing slides based on “AI Scientists as Engines of Discovery”. ArXiv: 2606.22859

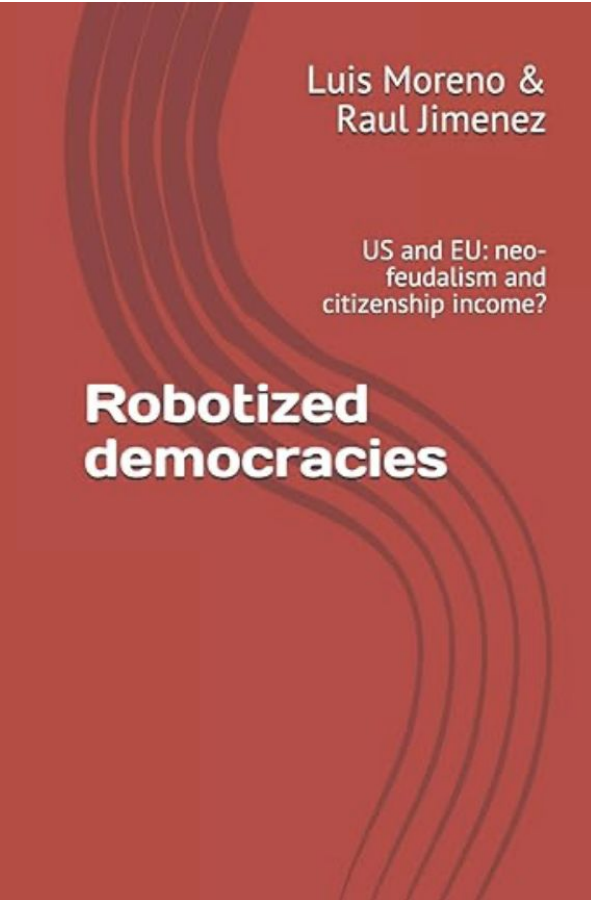


Thesis for debate: the key transition is not faster calculation, but AI systems entering the cognitive loop of science — proposing, criticizing, testing and revising models.

Question to the panel: what should cosmology automate, and what must remain a human scientific responsibility?

I wrote a book (10 years ago) about how to automate our democracies

And **just** wrote a **textbook in TWO volumes** on how to use ML to solve (partial and nonlinear) differential equations and inverse problems .



A proof of an identity for the critical exponents of jamming

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Abstract

Within the full replica-symmetry-breaking (fullRSB) solution of dense hard spheres in infinite dimension, Charbonneau, Kurchan, Parisi, Urbani, and Zamponi (CKPUZ; J. Stat. Mech. P10009, 2014) introduced three critical exponents a, b, c governing the matching region of the fullRSB profile near the jamming transition. These exponents satisfy two scaling relations. The first, $b = (1+c)/2$, was established analytically by the diffusion-drift balance in the scaling ansatz. The second, $a + b = 1$, was observed numerically to arbitrary precision but could not be proven. The exponents a, b, c of the scaling fullRSB ansatz are related to the physical exponents α, θ, κ that control the gap, force, and overlap distributions by the relations $\alpha = a/b$, $\theta = (c-a)/(b-c)$, $\kappa = c+1$. Crucially, the relation $a + b = 1$ yields the scaling relations $\alpha = 1/(2+\theta)$ and $\kappa = 2 - 2/(3+\theta)$ predicted on independent grounds by the mechanical-marginal-stability arguments of Wyart and collaborators. Here, we give an analytic proof of the identity $a + b = 1$ from the scaling fullRSB equations. The proof was obtained through interaction with Claude (Sonnet 4.6 and Opus 4.7) and verified by us.

Introduction and motivation

Jamming, marginal stability, and critical exponents

ıorphous packings of hard or soft spheres exhibit a non-equilibrium "jamming" transition the density above which mechanical rigidity sets in [1, 2, 3, 4]. Near this transition, a

#giorgioparisi #ia #intelligenzaartificiale #claude #scienza #tecnica #llm #retineurali #sapienzauniversitàdiroma | Enrico De Santis, Ph.D.

Eccoci qua: Giorgio Parisi, premio Nobel per la Fisica 2021 e Professore alla Sapienza, pubblica uno studio scientifico con una dimostrazione matematica sviluppata con l'IA, dichiarandolo apertamente.
Nell'abstract si legge:
«La prova è stata ottenuta tramite interazione con Claude (Sonetto 4.6 e Opus 4.7) e...

linkedin.com

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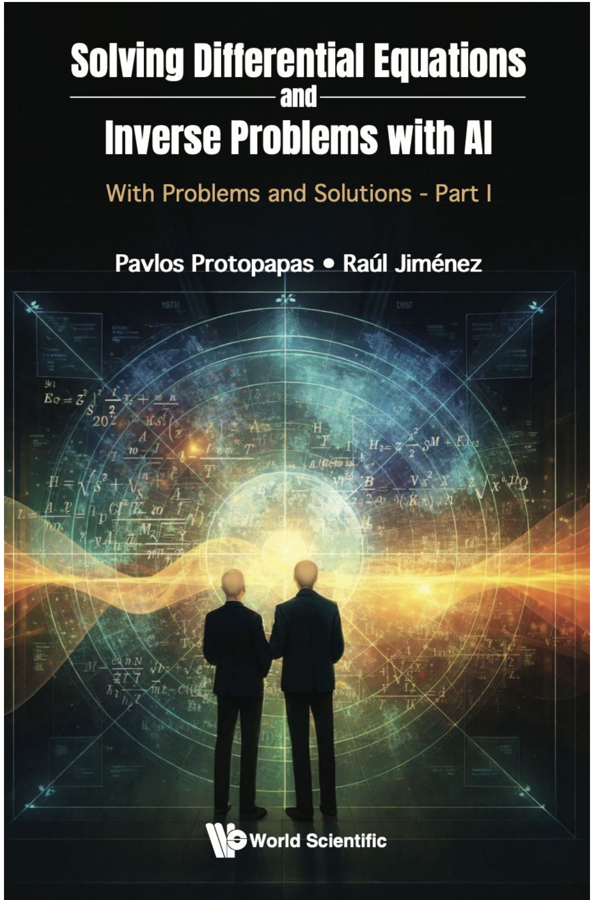

Cosmology is an ideal laboratory for AI science

Modern cosmology sits at the intersection of enormous data, expensive simulations, theory landscapes, and delicate systematics. This makes it unusually exposed to both the promise and risks of agentic AI.

Data scale

CMB, LSS, lensing, SNe, gravitational waves and multi-survey cross-correlations.

Simulation scale

N-body, hydro, emulators, pipelines and likelihoods that already exceed single-human mastery.

Model space

Inflation, dark energy, modified gravity, systematics and nuisance sectors form vast hypothesis spaces.

Inference risk

Plausible outputs are dangerous: a wrong prior, bug or citation can look scientifically fluent.

Panel prompt: can AI enlarge the cosmological model space without turning interpretation into a high-volume pattern-matching exercise?

From chatbot to algorithmic research group

The most consequential systems will not be isolated prompts. They will be coordinated research workflows with agents that generate, criticize, verify and orchestrate.

Generation agent
proposes models

→

Critic agent
finds failure modes

→

Verification agent
tests against data

→

Controller agent
orchestrates workflow

Denario (https://astropilot-ai.github.io/DenarioPaperPage/) illustrates this architecture: literature exploration, code execution, data analysis and iterative hypothesis testing coordinated by a reasoning layer.

SCAN ME

Cosmology use case: search theory landscapes, stress-test assumptions, reproduce pipelines and propose discriminating observations.

The human scientist does not disappear

AI can industrialize exploration of the adjacent possible. The human task becomes more demanding: choose what matters, build new conceptual frames, and remain accountable for meaning.

Machines contribute: speed, scale, memory, code, simulation, systematic model traversal
Humans contribute: problem framing, taste, conceptual synthesis, responsibility, interpretation
The danger: paper production without understanding; output volume mistaken for discovery

Panel prompt: if AI can do much of “normal science”, how should we train young cosmologists to create new questions rather than only operate pipelines?

Items for panel debate?

The article's claim is not "slow AI down". It is: make AI-augmented science verifiable, accountable and genuinely discovery-oriented.

Provenance	Should AI-assisted cosmology papers ship machine-readable records of prompts, code, data, and model decisions?
Verification	Can journals and collaborations build automated checks for code, statistics, citations and likelihood pipelines?
Novelty	How do we prevent shared models from homogenizing methods and rewarding the familiar?
Human loop	Which actions require explicit human authorization: experiments, releases, self-modifying agents, public claims?
Credit	How do we acknowledge AI contributions without giving AI authorship or evading human responsibility?

Closing question: Who will be in the driver's seat — and what kind of scientific enterprise do we want AI to amplify?